Verification matrix

Source: 04_architecture/verification_matrix.md; last changed 2026-10-15.

Two parts: the **current** results (final pre-submission run, 2026-10-15) and, below them, the dated **history** of the Phase 1 clean-state run, kept as a record. Counts in the history are Phase 1 counts and are superseded by this section.

Current results (final pre-submission run, 2026-10-15)

Run on the committed tree, in this order, one suite at a time.

Command	Exit	Result
`python -m unittest discover -s tests`	0	**185 run, OK**
`python 03_evals/run_golden_evals.py`	0	**33/33**
`python 03_evals/mutation_test.py`	0	**19/19 mutants caught**, 0 survived
`python 03_evals/robustness/fault_injection.py`	0	**38/38 faults behaved as required**; source workbook hash unchanged
`git ls-files -z | xargs -0 python scripts/scan_secrets.py`	0	175 text files scanned, 0 findings
`python scripts/check_links.py`	0	0 broken links (count in the final run record)

Not re-run in this pass: the grader on the three commentary texts, the pipeline for abc/xyz, the loader dry run and the live QuickBooks checks. Their last results are in the history below and in `03_evals/FINAL_RUN.md`.

Brief → proof → status (current)

#	Brief line	Proof (command / file)	Today	Status
1.1	Connect securely and reliably	`01_connection/connection_decision.md`; `capture_dated.py` check (c)	71 read-only tools; write tools not registered (`register-tool.ts`)	Done
1.2	Demonstrate it working against the sandbox	`qbo_test_connection.py abc --test`; `evidence/quickbooks/2026-10-14/`	CompanyInfo, P&L and BS all PASS, 04:34–04:36 UTC	Done
1.3	Document chosen / rejected / why	`connection_decision.md`	Doc review only	Done
1.4	Team hosting	`connection_decision.md` "Team hosting"	Design, labelled Proposed	Partial
1.5	Credential handling	`.claude/hooks/guard_secrets.py`, `scan_secrets.py`, `tests/test_evidence_redaction.py`	Scans clean; redaction residual check PASS	Done
2.1	Revenue, margins, opex by category	`src/abc_workbook_adapter.py`; G01–G03, G09–G13 (33 cases in total)	33/33	Done

#	Brief line	Proof (command / file)	Today	Status
2.2	MoM and TTM	`src/financial_analysis.py` ; golden TTM cases	33/33	Done
2.3	Commentary flagging movements	`grade_commentary.py` on engine, reviewed and skill text	3 PASS; flawed control FAIL	Done
2.4	For a human reviewer to finish	`src/monthly_review.py build …` → `monthly_review_2026-06.xlsx`	Rebuilt, APPROVED; same cell values as `02_skill/sample_outputs/`	Done
2.5	Maintainable, extensible	`02_skill/README.md`, `tests/`	185 tests OK	Done
2.6	Another company without a rewrite	`run_client.py xyz`, `run_pipeline.py xyz 2026-05`, `tests/test_client_isolation.py`	Both exit 0	Partial (wide-sheet layout only)
2.S	Stretch: load workbook into sandbox, run end to end	`load_workbook_to_sandbox.py --dry-run`; `--compare` vs today's live P&L	Dry run OK. Live P&L has **no** `Workbook Load` accounts (income 0.00 Jan–Apr 2026); `--compare` exits 1	Not done
3.1	≥5 golden cases	`run_golden_evals.py`	33/33	Done
3.2	Grading approach	`03_evals/GRADING.md`, `grade_commentary.py`	Grader separates good and flawed text	Done
3.3	Regression detection	`mutation_test.py`	19/19 caught	Done
3.4	Commentary quality beyond numbers	`GRADING.md`	Tone and causes not code-graded	Partial
4.1	Structure, files, skills, memory	`04_architecture/architecture.md`	Doc review only	Done
4.2	Hosting	`architecture.md` "Hosting"	Doc review only	Done
4.3	Ten clients, data separation	`tests/test_client_isolation.py`	In unit run, OK	Partial (two clients; OS/credential separation Proposed)
D1	Working connection	`01_connection/`	Checks a–d PASS; MCP server is an untracked clone	Partial
D2	Skill files	`.claude/skills/`, `.claude/commands/monthly-analysis.md`	Saved skill output graded PASS; the skill was **not** re-run live today	Done
D3	Eval cases and grading	`03_evals/`	As above	Done
D4	One-page architecture note	`architecture.md`	Doc review only	Done
R1	Demo connection and analysis	`qbo_test_connection.py --test`, `run_pipeline.py`	Both pass today	Done

#	Brief line	Proof (command / file)	Today	Status
R2	One small modification	`evals_caught.md` "Rehearsed modification", commit `315a473`	Mutation run now covers G18	Done
R3	Why structured this way	`architecture.md`, `workflow.md`	Doc review only	Done
R4	What the evals caught	`03_evals/evals_caught.md`	Today's fixes listed above	Done
R5	What breaks first at ten clients	`architecture.md` "What breaks first"	Doc review only	Done
G1	Sandbox and supplied workbook only	`CLAUDE.md`, `clients/xyz/` (synthetic)	Workbook hash unchanged	Done
G2	Explain every choice as your own	Docs	Depends on the presenter	Partial

History: Phase 1 clean-state run (2026-10-14)

> **These are Phase 1 counts** (103 unit tests, 18/18 golden). Later phases added G19–G22, the 32-fault robustness suite and workflow-failure tests; see the current results above.

Every line of the brief, mapped to the command or file that proves it and the result of the Phase 1 run on 2026-10-14, from a clean `outputs/`. The old `outputs/` was moved aside first and everything was regenerated. Authored inputs that no script produces were then restored: reviewed and skill commentary, reviewer verdicts, the reference commentary, the 3-month reconciliation and earlier run manifests. Audit logs were restored with their history first and today's entries after. Status wording is the same as `brief_traceability.md`.

Runs (Phase 1)

Command	Exit	Key result
`python src/run_client.py abc` / `xyz`	0 / 0	abc: 18 months, 0 warnings. xyz: 14 months, 1 warning (no Sales & Marketing accounts; reported as not available)
`python scripts/run_pipeline.py abc 2026-06` / `xyz 2026-05`	0 / 0	`success`; intake, calculate and grade stages done
`python 03_evals/run_golden_evals.py`	0	18/18
`python 03_evals/mutation_test.py`	0	13/13 mutants caught, 0 survived. First run since the G18 margin-drop change; G18 and `test_gross_margin_drop_flags` now also catch the margin-denominator mutant
`python 03_evals/grade_commentary.py` (engine)	0	PASS
`… --commentary outputs/abc/commentary_2026-06.md` (reviewed)	0	PASS: 27/27 numbers, 3/3 movements, cash risk covered

Command	Exit	Key result
`… --commentary outputs/skill_commentary.md` (saved skill output)	0	PASS: 36/36, 3/3, cash risk covered
`… --commentary outputs/abc/commentary_2026-06_flawed_control.md` (negative control)	1	FAIL as intended: 11/12, 2/3, cash risk missing
`python -m unittest discover -s tests`	0	103 run, OK. The first clean-state run errored; see Fixes
`python scripts/scan_secrets.py .`	1	28 findings, **all** inside the untracked, git-ignored `quickbooks-online-mcp-server/` clone (Intuit's code and tests; its `.env` is listed by name only, never opened). 0 outside it
`git ls-files -z | xargs -0 python scripts/scan_secrets.py`	0	145 text files scanned, 0 findings. Before today's fix the same command scanned **0** files; see Fixes
`python 01_connection/load_workbook_to_sandbox.py --dry-run`	0	18-entry plan, nothing created
Workbook SHA-256 before / after	—	`75da8d679a89551f…` both times

Live sandbox checks (read-only): see `01_connection/evidence/quickbooks/2026-10-14/README.md`. (a) CompanyInfo PASS. (b) P&L Jan–Jun 2026 monthly and Balance Sheet at 30 Jun 2026 PASS; the BS balances at 5,488.75. (c) 71 tools registered, none of them create/update/delete, PASS. (d) A missing server folder gives an actionable message with exit 1, PASS.

Fixes made in Phase 1

Found by	Defect	Fix	Test
Clean-state test run	`tests/test_ui_numbers.py` crashed with `FileNotFoundError` when `outputs/commentary_grade.json` did not exist yet (test-order dependency)	Missing global eval files are left out of the allowed-token set. That can only make the test stricter	Re-run without the file: 15 UI tests OK
Clean-state regeneration	Committed `outputs/abc/review_packet_2026-06.json` was stale: it lacked `gross_margin_flags` from the G18 change	Regenerated by `run_pipeline.py`	Golden G18, unit tests
Tracked-file secret scan	`scan_secrets.py` treated every argument as a folder: given files it scanned 0 and reported "0 findings". A misspelt path also passed	Files are scanned directly; a missing path exits with an error	`test_file_paths_are_scanned_and_missing_paths_fail`
Step 3d	`qbo_test_connection.py --test` exited 0 when the test failed, and the review UI (`actions._run`) gates on the exit code	Exits 1 on a failed test	Live run: exit 1 for a missing server, exit 0 when connected
Step 3d	No way to point the test at another server folder; the missing-server message named a fixed folder	`QBO_MCP_SERVER_DIR` override; the message names the folder it looked in	Live run above

Independent falsification check

A subagent that had not seen this work got only the brief and the repo, and tried to falsify 5 claims in `README.md` and `brief_traceability.md`. All 5 held: - unit 103 OK, golden 18/18; - the golden cases by source; - golden expected values for G01, G02, G03, G06 and G09, recomputed from workbook cells; - 13 mutants; - `xyz` runs on config only.

It found two more problems: - **"The note is 38 lines" was misleading.** The note is 614 words, with long lines. Reworded in `brief_traceability.md`. - **A unit test wrote into the tracked `outputs/`.** `test_qbo_load` ran `--apply` without an env file, and the loader wrote `outputs/abc/qbo_load_plan.json` before it stopped. Fixed: the loader now stops before any side effect, and the test runs against a temporary output root and asserts nothing was written. The test fails without the fix and passes with it.

Not checked by anyone: G10–G18 cell by cell, and whether `architecture.md` fits on one printed page.

"Doc review only" means the line is met by a document's content; no command can prove it.

Robustness report: fault injection

Source: 03_evals/robustness/ROBUSTNESS_REPORT.md; last changed 2026-10-15.

Generated by `python 03_evals/robustness/fault_injection.py` (do not edit by hand). Each fault is applied to a COPY of the ABC workbook in a temp folder, and both entry points that produce numbers run on that copy: `scripts/run_pipeline.py` (with `--clients-root/--outputs-root` pointing at the temp folder) and `monthly_review.prepare`, step 1 of the `/monthly-analysis` skill. A fault passes only if both paths behave as required.

Source workbook SHA-256 before `75da8d679a89551f…`, after `75da8d679a89551f…` (unchanged).

Required behaviour, fixed before running: **STOP** = intake refuses with a message naming the cell, sheet or cause, and no review packet is written; **WARN** = run continues and the warning is in the review packet (Open Questions in the sign-off pack) and in the analysis workbook's Data Quality sheet; **HANDLE** = run continues, no new warning, every review-packet number and every account amount in the analysis workbook (`Account Values` sheet) equals the baseline. **SILENT-WRONG** (success, different numbers, no warning) must never occur.

Result: 38/38 pass; SILENT-WRONG: 0.

Stricter than declared: a fault declared WARN that the Balance Sheet tie-out refuses (STOP) before the warning could happen. Passes because nothing wrong is accepted; the declarations were not edited. Before the tie-out existed (commit c8421e4) these ran as WARN. Rule: an expectation only moves toward stricter (HANDLE, WARN, STOP), never looser; no row moved looser. Rows moved to STOP: F02, F13, F16, F17, F21, F27. Per-fault reasons: `03_evals/evals_caught.md`.

WARN-path variants (F02b, F13b, F16b, F17b, F21b, F27b): the same six faults with the Balance Sheet restated so the tie-out passes (Current Year Earnings reset to the mutated year-to-date net income, Retained Earnings takes the opposite change, so the sheet still balances and cash is unchanged). Required, declared in code before running: WARN end to end, i.e. the warning is in `summary.json`, the review packet, the analysis workbook's Data Quality sheet and the reviewer Excel's Open Questions (built by `monthly_review.build` over that run's packet), and every detail-account amount other than the accounts the fault itself changes equals the baseline, as does the packet's cash. The originals stay as STOP cases, so both the refusal and the warning path are covered.

ID	Family	Fault	Required	Pipeline	Skill path	Pass	Message shown (pipeline)
F01	Structure	Missing month column: prior month (May 2026) deleted	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Income Statement: no column for 2026-05 (the prior month of 2026-06) in header row 5; months found: 2025-01 to 2026-06.
F02	Structure	Missing month column: Sep 2025 deleted mid-history	WARN	STOP	STOP	Pass (stricter)	ValueError: Intake checks failed (details in intake_2026-06.json): Current Year Earnings = year-to-date net income (every month) at Balance Sheet!L63, Balance Sheet!K63, Balance Sheet!J63 (Fix: check that the Balance Sheet and Income Statement are from the same export; 2025-10: balance sheet -3,662,740.34 vs YTD net income -3,302,388.89; 2025-11: balance sheet -3,949,650.38 vs YTD net income -3,58
F03	Structure	Duplicated month (Jun 2026 column copied next to itself)	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Income Statement: month 2026-06 appears in more than one column (C5, D5).
F04	Structure	Months out of order (columns interleaved)	HANDLE	HANDLE	HANDLE	Pass	Run succeeded; every review-packet number and every account amount equals the baseline.
F05	Structure	Extra blank column inside the month columns	STOP	STOP	STOP	Pass	ValueError: Intake checks failed (details in intake_2026-06.json): Month columns contiguous at Income Statement!C5:U5 (18 month columns)
F06	Structure	Extra blank rows inside the data	HANDLE	HANDLE	HANDLE	Pass	Run succeeded; every review-packet number and every account amount equals the baseline.

ID	Family	Fault	Required	Pipeline	Skill path	Pass	Message shown (pipeline)
F07	Structure	Header row shifted down 3 rows	HANDLE	HANDLE	HANDLE	Pass	Run succeeded; every review-packet number and every account amount equals the baseline.
F08	Structure	Income Statement sheet renamed to 'P&L'	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Sheet 'Income Statement' not found in book.xlsx; sheets present: ['P&L', 'Balance Sheet'].
F09	Structure	Balance Sheet missing	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Sheet 'Balance Sheet' not found in book.xlsx; sheets present: ['Income Statement'].
F10	Structure	Extra unknown sheet	HANDLE	HANDLE	HANDLE	Pass	Run succeeded; every review-packet number and every account amount equals the baseline.
F11	Structure	Merged header cells (Jun and May 2026)	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Income Statement: header cells are merged (C5:D5); unmerge them so each month has its own header. | Income Statement: no column for 2026-05 (the prior month of 2026-06) in header row 5; months found: 2025-01 to 2026-06.
F12	Values	Text 'n/a' in a numeric cell (Usage Fees, Jun 2026)	STOP	STOP	STOP	Pass	ValueError: Intake checks failed (details in intake_2026-06.json): Every amount is a number at Income Statement!C11 (Usage Fees Jun 2026: text, not a number: 'n/a') | Total Income foots to its lines (every month) at Income Statement!C14 (2026-06: lines - total = -166,781.50)
F13	Values	Empty cell inside a series (Bonus, Feb 2026; totals treat it as zero)	WARN	STOP	STOP	Pass (stricter)	ValueError: Intake checks failed (details in intake_2026-06.json): Current Year Earnings = year-to-date net income (every month) at Balance Sheet!H63, Balance Sheet!G63, Balance Sheet!F63, Balance Sheet!E63, Balance Sheet!D63 (Fix: check that the Balance Sheet and Income Statement are from the same export; 2026-02: balance sheet -720,890.81 vs YTD net income -710,210.81; 2026-03: balance sheet -99
F14	Values	Amount as text '$1,234.56' (Platform Revenue, Jun 2026)	HANDLE	HANDLE	HANDLE	Pass	Run succeeded; every review-packet number and every account amount equals the baseline.
F15	Values	Negative as text '(1,234.56)' (Bad Debts, Jun 2026)	HANDLE	HANDLE	HANDLE	Pass	Run succeeded; every review-packet number and every account amount equals the baseline.
F16	Values	Extreme outlier x1000 (Usage Fees, Jun 2026, totals consistent)	WARN	STOP	STOP	Pass (stricter)	ValueError: Intake checks failed (details in intake_2026-06.json): Current Year Earnings = year-to-date net income (every month) at Balance Sheet!D63 (Fix: check that the Balance Sheet and Income Statement are from the same export; 2026-06: balance sheet -1,987,916.44 vs YTD net income 164,626,802.06)
F17	Values	Zero total revenue in Jun 2026 (division by zero)	WARN	STOP	STOP	Pass (stricter)	ValueError: Intake checks failed (details in intake_2026-06.json): Current Year Earnings = year-to-date net income (every month) at Balance Sheet!D63 (Fix: check that the Balance Sheet and Income Statement are from the same export; 2026-06: balance sheet -1,987,916.44 vs YTD net income -2,183,814.60)
F18	Values	Text 'NaN' in a numeric cell (Customer Success - Payroll benefits, Jun 2026)	STOP	STOP	STOP	Pass	ValueError: Intake checks failed (details in intake_2026-06.json): Every amount is a number at Income Statement!C17 (Customer Success - Payroll benefits Jun 2026: text, not a number: 'NaN') | Total Cost of Goods Sold foots to its lines (every month) at Income Statement!C24 (2026-06: lines - total = -831.37)
F19	Values	Excel error value '#DIV/0!' in a numeric cell (Bonus, Jun 2026)	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Income Statement: 1 cells hold an Excel error value (C33 #DIV/0!). Fix the formula and re-run.
F20	Labels	Account renamed (Bonus -> Bonuses)	WARN	WARN	WARN	Pass	Unmapped OPEX account (reported under 'Unmapped'): Bonuses
F21	Labels	New unmapped account (Crypto Mining Costs, totals consistent)	WARN	STOP	STOP	Pass (stricter)	ValueError: Intake checks failed (details in intake_2026-06.json): Current Year Earnings = year-to-date net income (every month) at Balance Sheet!U63, Balance Sheet!T63, Balance Sheet!S63, Balance Sheet!R63, Balance Sheet!Q63 (+13 more) (Fix: check that the Balance Sheet and Income Statement are from the same export; 2025-01: balance sheet -382,905.03 vs YTD net income -383,005.03; 2025-02: balanc
F22	Labels	Trailing spaces on every label	HANDLE	HANDLE	HANDLE	Pass	Run succeeded; every review-packet number and every account amount equals the baseline.
F23	Labels	Unicode look-alike label (Bonus with a Greek omicron)	WARN	WARN	WARN	Pass	Account label has non-ASCII characters (possible look-alike or typo): 'Bonus'

ID	Family	Fault	Required	Pipeline	Skill path	Pass	Message shown (pipeline)
F24	Labels	Missing section total row (Total Cost of Goods Sold deleted)	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Income Statement: total row 'Total Cost of Goods Sold' (cost_of_goods_sold) not found. It is required; check for a renamed or deleted row. | Income Statement: section total row 'Total Cost of Goods Sold' not found; the cost of goods sold section cannot be closed or footed.
F25	Labels	Missing block subtotal row (Total Marketing deleted)	HANDLE	HANDLE	HANDLE	Pass	Run succeeded; every review-packet number and every account amount equals the baseline.
F26	Labels	Subtotal does not foot (+1,000 on Usage Fees, Mar 2026)	STOP	STOP	STOP	Pass	ValueError: Intake checks failed (details in intake_2026-06.json): Total Income foots to its lines (every month) at Income Statement!F14 (2026-03: lines - total = 1,000.00)
F27	Coverage	Only 11 months (Aug 2025 - Jun 2026)	WARN	STOP	STOP	Pass (stricter)	ValueError: Intake checks failed (details in intake_2026-06.json): Current Year Earnings = year-to-date net income (every month) at Balance Sheet!N63, Balance Sheet!M63, Balance Sheet!L63, Balance Sheet!K63, Balance Sheet!J63 (Fix: check that the Balance Sheet and Income Statement are from the same export; 2025-08: balance sheet -2,955,994.67 vs YTD net income -316,771.89; 2025-09: balance sheet -
F28	Coverage	Saved with formulas but no cached values	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Income Statement: 144 formula cells have no saved value (C14, D14, E14, F14, G14 (+139 more)). The file was saved without calculating; open it in Excel, save, and re-run. | Balance Sheet: 180 formula cells have no saved value (D15, E15, F15, G15, H15 (+175 more)). The file was saved without calculating; open it in Excel, save, and re-run.
F29	Coverage	Corrupt file (not a zip)	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Not a valid .xlsx workbook (the file is not a zip package): book.xlsx. Re-export it from the source system.
F30	Coverage	Truncated .xlsx (zip header, cut off)	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Cannot open book.xlsx as an .xlsx workbook (BadZipFile). Re-export it from the source system.
F31	Coverage	File open and locked by another program	STOP	STOP	STOP	Pass	ValueError: Intake refused the workbook: Workbook is open or locked by another program: \clients\abc\book.xlsx. Close it (for example in Excel) and re-run.
F32	Values	Float inf written into a numeric cell (Bonus, Jun 2026; lands as an empty cell)	STOP	STOP	STOP	Pass	ValueError: Intake checks failed (details in intake_2026-06.json): Total Operating Expenses foots to its lines (every month) at Income Statement!C103 (2026-06: lines - total = -10,889.00)
F02b	Structure	Missing month column: Sep 2025 deleted mid-history [Balance Sheet restated to tie out]	WARN	WARN	WARN	Pass	Months missing from the source: Sep 2025. TTM is withheld for windows with gaps, and no month-over-month change is reported across a gap.
F13b	Values	Empty cell inside a series (Bonus, Feb 2026; totals treat it as zero) [Balance Sheet restated to tie out]	WARN	WARN	WARN	Pass	1 blank cell(s) inside rows that have values, treated as missing (not zero): Income Statement!G33 (Bonus, Feb 2026)
F16b	Values	Extreme outlier x1000 (Usage Fees, Jun 2026, totals consistent) [Balance Sheet restated to tie out]	WARN	WARN	WARN	Pass	Possible outlier: Usage Fees in Jun 2026 is 166,781,500.00, over 100x its median month (140,396.16). Check Income Statement!C11.
F17b	Values	Zero total revenue in Jun 2026 (division by zero) [Balance Sheet restated to tie out]	WARN	WARN	WARN	Pass	Revenue is zero in Jun 2026; margins for that month are reported as not available.
F21b	Labels	New unmapped account (Crypto Mining Costs, totals consistent) [Balance Sheet restated to tie out]	WARN	WARN	WARN	Pass	Unmapped OPEX account (reported under 'Unmapped'): Crypto Mining Costs

ID	Family	Fault	Required	Pipeline	Skill path	Pass	Message shown (pipeline)
F27b	Coverage	Only 11 months (Aug 2025 - Jun 2026) [Balance Sheet restated to tie out]	WARN	WARN	WARN	Pass	TTM not available: 11 months supplied, 12 consecutive required.